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Group Assignment

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Name	**Student ID**
HAZEL FONG HWEI YEE	0385678
RACHEL ANG DONG JIN	0383084
CHIN YUAN KANG	0384858

Declaration

I certify that this assignment is entirely my own work, except where I have given fully documented references to the work of others, and that the material contained in this assignment has not previously been submitted for assessment in any other formal course of study.

Student's Signature(s)	**Date**
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Comments:

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1.0 Introduction

This report focuses on helping a small to medium-sized company to design a smooth and secure network. The company is a 3-storey office building, and it is estimated to have around 100 employees distributed across all floors. The company mainly uses the internet for video conferencing which concerns on real time data being transferred and the use of cloud-based applications that is required to be connected to the internet.

This report will help the company to decide the correct wired and wireless connectivity, network segmentation implementation, ensuring that the network has high availability, and using the appropriate error detection mechanisms.

2.0 Network Diagram

The following network diagram will show the relationship and connections within the network connecting various devices.

2.1 Network Topology Diagram

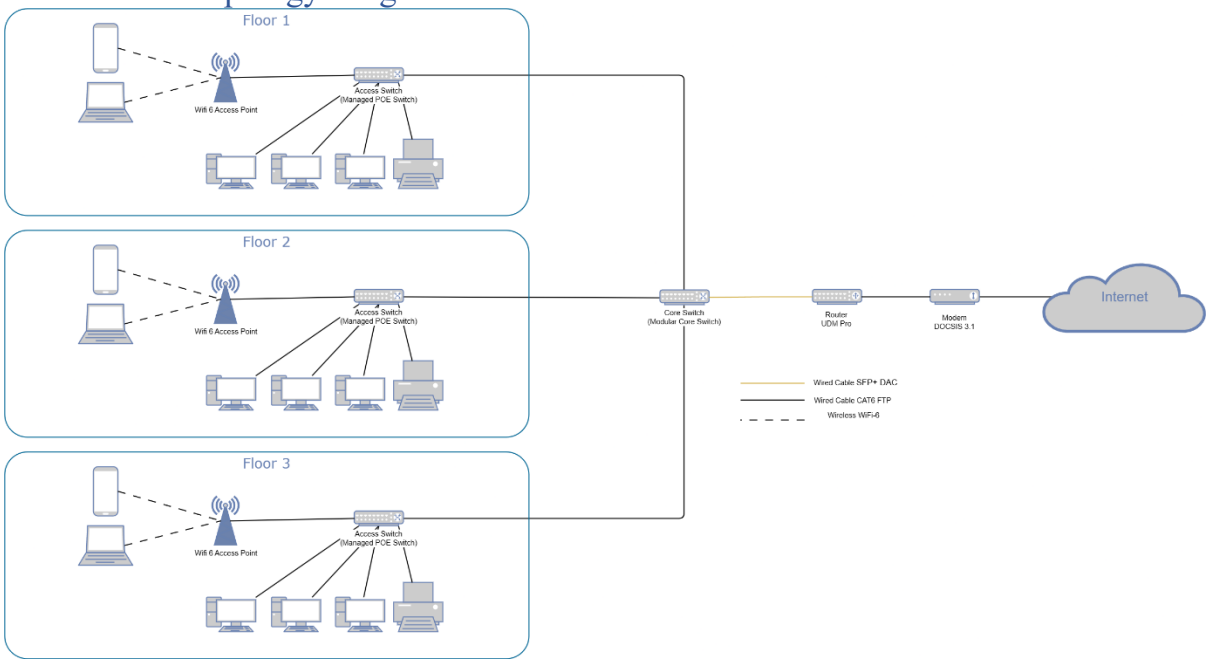


Figure 1 Network Topology Diagram

The planning of the network topology diagram starts with a CAT6 FTP cable that links the internet service and modem together so the company can get access to the internet. Then the modem will be connected to the router using another CAT6 FTP cable. Next, the router and core switch is connected using a different cable, which is 10 G Direct Attach Copper to provide high-speed connection. After that, the core switch will connect to a managed Power over Ethernet network switch on each floor using CAT6 FTP. And every PoE switch will link to one WIFI access point and multiple wired devices like desktops and printers through CAT6 FTP. Personal devices such as mobile phones and laptops that are wireless can connect to the access point to access the local network or internet. This is a hierarchical star topology that will help improve the company network.

2.2 Recommendations

In this section, components for hardware and software are recommended according to the scenario and network topology diagram.

2.2.1 Hardware

2.2.1.1 Wired Connection

2.2.1.1.1 Network Cable

In a network system, it is crucial to have network cables as it is the medium of transmitting data between different devices. Without network cable, data would not be able to transfer between devices during wired connection. There are multiple types of network cable such as coaxial cable, twisted pair cable, and fiber optic cable (Rogers, 2026) In this scenario, there are two cable that is recommended which is twisted pair cable category Cat 6 and the type Foiled Twisted Pair (FTP) and a 10 G Small Form-factor Pluggable (SFP)+ Direct Attach Copper (DAC) are recommended. The figure below shows the twisted pair cable with the category of Cat 6 and the type of FTP.



Figure 2 Cat 6 FTP

The reason for choosing twisted-pair cables is that they are more cost-effective compared to fiber optic cables (Rogers, 2026). This happens as the process of manufacturing fiber optic cables is complex due to the material being raw and it requires high purity from the material for light signals to be transferred (Fmyly, 2025).

On the other hand, Coaxial cables are not recommended because they have limited bandwidth compared to the twisted-pair cable (Rogers, 2026). By having limited bandwidth, the maximum amount of data that can be transmitted would also be limited (Pioth, 2026). This would result in slow data transmission and seriously affect the company since the company has high internet usage and requires real-time data transmission.

So, it is recommended to use a twisted-pair cable. Twisted-pair cables have different types of categories, and each category supports different max data speed and max bandwidth (Product Knowledge, 2023). It is recommended to purchase cable types that offer high speed support as high-speed support means the rates of data being transferred are higher and will prevent bottlenecks (Fycables, 2025). This is important as video conference requires high data transfer rates for data to be transferred smoothly.

To be more specific, Cat6 FTP cable is recommended as it supports 10Gbps speed over short distance and FTP will have a metallic or foil shielding layer to reduce interference to maintain high-quality video conferencing and reduce the rate of data being corrupted. (Universal Networks, n.d.)

As for SFP+ Direct Attach Copper (DAC), it is a copper-based cable that offers high speed connectivity. It is typically used to interconnect network switches, servers, or storage devices. The figure below shows the SFP+ Direct Attach Copper (DAC) that is recommended.



Figure 3 10G DAC SFP+ Cable

Another recommended cable is the SFP+ Direct Attach Copper (DAC). They are copper-based cables that offer high speed connectivity. It is typically used to interconnect network switches, servers, or storage devices. Their ability to significantly increase the bandwidth between two devices plays an important role because it allows the data transfer to flow fast and smoothly between the two devices. This will result in better user experience, higher resolution on video streams, better performance on file servers or virtual machines, and really improve the performance of any device that needs high bandwidth, fast, and clear signal.

There are two types of DAC, passive DAC and active DAC. For this scenario, it is a better option to use passive DAC cable. The reason for using passive DAC instead of active DAC is commonly the core switch and router would be placed on the same rack, so the former cable has a lower cost, and it does not need for a longer reach and 1 to 3 meters is enough (Network Switches, 2025).

To suggest one of the most convenient and cost-effective options for 10G networks to the company is SFP+ DAC Cable. They're easy to install and use, are low latency and cost-effective, and are designed to plug and play between servers, storage, and switches within or across racks. (Network Switches, 2025)

2.2.1.1.3 Core Switch

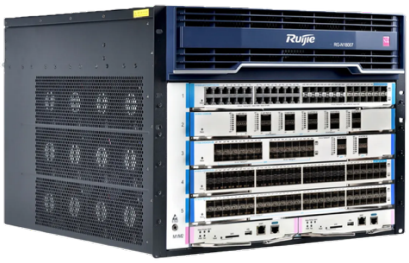


Figure 4 Modular Core Switch

Core switch is the backbone of a network design as it helps to handle massive volume of traffic with ultra-low latency and maximum reliability (Gong, n.d.). This happens as it centralizes multiple aggregation of switches and implement LAN routing (Rujie, 2023). These characteristics are important for the company as the company uses high internet due to regular video conferencing and cloud applications (Gong, n.d.). Therefore, modular core switch is recommended in this proposed network. This happens as the cost for modular core switch and the complexity of it is average. (Powell, 2025) This means the cost of it would not be too high and it does not require too much learning in how to use or implement the modular core switch.

2.2.1.1.4 Routers



Figure 5 Router UDM Pro

The function of a router in networking is to control the flow of data and forward it to the correct IP address whilst allowing multiple clients to use the same internet connection. Therefore, a router plays an important role of managing the traffic of the network. There exist two types of routers, wired and wireless. (Cloudfare, n.d.)

In this scenario, it is recommended to use UDM Pro (UniFi Dream Machine Pro) for the router. UDM Pro is a wired router combines a powerful router, a firewall, a network controller, and an NVR, which stand for network video recording support into one device. This device will be the main component of the network; it can support over 100 UniFi devices, more than 1000 client devices, and manage around 24 standard HD cameras, 14 2K resolution cameras, or 8 4K high-definition cameras. It is more than capable to support 100 employees across three-storey floor.

UDM Pro has eight 1GB RJ45 LAN ports, one 1Gb Ethernet WAN port and two 10G SFP+ ports. The only downside is that none of these ports provide PoE (Power over Ethernet). A PoE switch will be recommended to solve this problem. It is capable of up to 3.5 Gbps throughput with IDS (Intrusion Detection System) and IPS (Intrusion Prevention System) enable. It also supports different type of VPN connectivity, SD-WAN features and network features. These features, specifically advanced Quality of Service (QoS) is important because it keeps video conferencing smooth even when there is high internet usage. (Microsoft, 2024) This is highly favourable for the company as it mainly uses video conferencing and wants the experience to be uninterrupted.

2.2.1.1.5 Switches



Figure 6 Managed PoE Switch

A network switch is a piece of hardware that allows network to be shared across multiple devices. It prevents the traffic for each device being interfered with other device's traffic (CISCO, n.d.). This hardware is important for the company as the internet of the company is high and workers in the office often use the network for video conferencing where it is important to prevent the traffic for each device from interfering with each other. This helps to ensure that the data to be sent and received will not be corrupted during the call due to the interference.

For the company, the wired connection switch recommended is to use a managed PoE switch. Switches with the type managed are recommended as managed switches offer more capacity compared to unmanaged switches where they offer limited capacity (CISCO, n.d.). This characteristic is important as the company has a 3-storey building and has 100 employees distributed across each floor. This causes the devices that will need to connect through switches such as printers; surveillance cameras will be a lot. Besides this, unmanaged switches are ideal to use when no segmentation is needed (Logicbus, 2025). This condition is against the requirement of the company as it requires segmentation of the network to separate each department.

The PoE switch, which is also known as Power over Ethernet switch, is recommended as the router proposed above does not provide electrical power and only transmits data. To allow devices to connect to the WIFI, access points are needed. PoE switches are responsible for providing power to the access points so that the access points can provide wireless connection to the devices in the building (Network Switches, 2026).

All in all, the Managed PoE switch is recommended to be the network switch in the diagram.

2.2.1.1.6 Modem



Figure 7 DOCSIS 3.1 Cable Modem

A modem is a device that helps to connect the network that is created by router to the internet (Chris Hoffman & Nick Lewis, 2023). There are many types of modems, but the modem recommended for this scenario would be cable modem. Cable modem with the version of DOCSIS 3.1 is recommended as it supports multi-gigabit plans. Its downstream offers 10Gbps max speed and 1.5 Gbps for maximum upstream speed (Martinez, 2025). This is critical for video calls and cloud backups as it has low latency upstream (Morgan, 2026). This suits the company as it has the requirement of high internet usage due to regular video conferencing and cloud applications.

2.2.1.2Wireless Connection

2.2.1.2.1 Wireless Access Point



Figure 8 - UniFi U6 Pro (Access Point)

Access point (AP) is a wireless network device that helps to cover areas with poor signals or no signals without using a physical cable. Plus, it increases the number of users that can connect to it. AP is a must have in the proposed network because it is more convenient, secure, and is

cost-efficient compared to using wires and cables to connect every computer or device. (Cisco, n.d.)

This access point UniFi U6 Pro is recommended for the scenario. It is connected by PoE (Power over Ethernet). This PoE connection sends data back and forth and helps to power the access point, this means a router or switch that supports PoE connection is crucial since this AP is implemented in the network. (Cisco, n.d.) To satisfy this requirement, the switch proposed above supports PoE connection.

AP is beneficial for the company because when the number of employees increase, it can provide high network scalability. This means that the company can easily expand by installing multiple access points. Other than that, the company can configure the AP to broadcast multiple WIFI network and map them into different VLAN. (Cisco, n.d.) This can help to control the internet, improve the company security and performance by separating employees and guests. (Randy, n.d.) There is a feature that if the access point is overloaded, the system will actively shift users from one access point to another depending on the capacity that is available, this is called high capacity load balancing.

UniFi U6 Pro official states that it can support 300 and more devices at the same time, and it can cover around 1500 square feet. (Unifi, n.d.) It is advised for the company to install one U6 Pro AP for each floor. The access point can be a ceiling or wall mount, and it is recommended to be placed at areas where people use their device, possible to be keep in sight from working desks. (Data Wire Solution, 2026)

UniFi U6 Pro is also waterproof, if the fire sprinkle got triggered unexpectedly due to false alarm, it can still function as it is, making it sturdy and reliable. (Henao, 2023) Overall, the UniFi U6 Pro is a well-thought through product recommended to the company satisfying the requirement.

2.2.2 Software

2.2.2.1 UniFi Application Suite

UniFi Application Suite is with the version 10.4.57 software is recommended as it contains a wide variety of applications and features in it.

In UniFi Application Suite there is UniFi Network where it plays a vital role in network-related aspects. This happens as the UniFi Network provides dashboard that contains real-time traffic, maps of visual topology and optimization tips (UniFi, n.d.) . By having these features, it allows the company to analyse on the traffic and enhance the performance of the network.

UniFi Protect is also included UniFi Application Suite. UniFi Protect is often used to see what is happening in the past and present. It also supports evidence if it is needed. Thus, alerts could also be received without constant noise. (Sistrun, 2026)

Beside this, UniFi Application Suite includes UniFi Talk where it is a locally hose Plug-and Play VoIP system. This includes admin features and advanced call routing (UniFi, n.d.). The advanced call routing system is important for the company as it helps to direct the callers to the right agent, team or department (Stone, 2024).

Furthermore, Unifi Access is also included in UniFi Application Suite. This UniFi Access is a platform design for access control system (Ng, 2025). This is important as it helps to verify the

identity and give or deny the access of a device from connecting to the network (Mahaldar, 2025). This helps to prevent unauthorized device from accessing the network to do malicious or suspicious activity.

Lastly, the UniFi Application Suite includes UniFi Connect where it helps to manage devices at scale (UniFi, n.d.). This happens as the administrator can monitor and configurate all device on a schedule without wasting much effort.

2.2.2.2 Intrusion Monitoring and Prevention System

Intrusion Detection System (IDS) is a mechanism that can monitor and find suspicious networking traffic and report it to the manager, and Intrusion Prevention System (IPS) is a mechanism that can eliminate suspicious traffic to ensure safe networking experience for the user. The configuration is easy to set up, and the steps are shown on the Ubiquiti Help Center website. This can help the company to save on extra operation fees or hiring a technician when they can configure the settings themselves. (UniFi, 2026)

2.3 IP Addressing Scheme

Class B: 128.155.0.0/16

Floor1						
Device Name	IP address	Subnet mask	Default gateway	Type of IP	VLAN	Subnetting
UDM Pro Floor1 Public Area	128.155.0.1	255.255.255.0	128.155.0.1	Static	none	none
Managed POE switch Floor 1 Public Area	128.155.0.2	255.255.255.0	128.155.0.1	Static	none	none
Devices From Sales Department	128.155.11.2	255.255.255.192	128.155.11.1	Static	VLAN 11	Subnet1: Network Address: 128.155.11.0 First Usable IP: 128.155.11.1 Last Usable IP: 128.155.11.62 Broadcast Address: 128.155.11.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.11.64 First Usable IP: 128.155.11.65 Last Usable IP: 128.155.11.126 Broadcast Address: 128.155.11.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.11.128 First Usable IP: 128.155.11.129 Last Usable IP: 128.155.11.190 Broadcast Address: 128.155.11.191 Subnet mask: 255.255.255.192

						Subnet4: Network Address: 128.155.11.192 First Usable IP: 128.155.11.193 Last Usable IP: 128.155.11.254 Broadcast Address: 128.155.11.255 Subnet mask: 255.255.255.192
Devices From Sales Department	128.155.12.2	255.255.255.192	128.155.12.1	Static	VLAN 12	Subnet1: Network Address: 128.155.12.0 First Usable IP: 128.155.12.1 Last Usable IP: 128.155.12.62 Broadcast Address: 128.155.12.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.12.64 First Usable IP: 128.155.12.65 Last Usable IP: 128.155.12.126 Broadcast Address: 128.155.12.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.12.128 First Usable IP: 128.155.12.129 Last Usable IP: 128.155.12.190 Broadcast Address: 128.155.12.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.12.192 First Usable IP: 128.155.12.193 Last Usable IP: 128.155.12.254

						Broadcast Address: 128.155.12.255 Subnet mask: 255.255.255.192
Devices From Client	128.155.13.2	255.255.255.192	128.155.13.1	Static	VLAN 13	Subnet1: Network Address: 128.155.13.0 First Usable IP: 128.155.13.1 Last Usable IP: 128.155.13.62 Broadcast Address: 128.155.13.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.13.64 First Usable IP: 128.155.13.65 Last Usable IP: 128.155.13.126 Broadcast Address: 128.155.13.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.13.128 First Usable IP: 128.155.13.129 Last Usable IP: 128.155.13.190 Broadcast Address: 128.155.13.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.13.192 First Usable IP: 128.155.13.193 Last Usable IP: 128.155.13.254 Broadcast Address: 128.155.13.255 Subnet mask: 255.255.255.192
One Printer	128.155.14.2	255.255.255.192	128.155.14.1	Static	VLAN 14	Subnet1: Network Address: 128.155.14.0

Public area Floor 1						First Usable IP: 128.155.14.1 Last Usable IP: 128.155.14.62 Broadcast Address: 128.155.14.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.14.64 First Usable IP: 128.155.14.65 Last Usable IP: 128.155.14.126 Broadcast Address: 128.155.14.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.14.128 First Usable IP: 128.155.14.129 Last Usable IP: 128.155.14.190 Broadcast Address: 128.155.14.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.14.192 First Usable IP: 128.155.14.193 Last Usable IP: 128.155.14.254 Broadcast Address: 128.155.14.255 Subnet mask: 255.255.255.192
U6pro Access Point At public area Floor1	128.155.10.2	255.255.255.128	128.155.10.1	Static	VLAN 10	Subnet1: Network Address: 128.155.10.0 First Usable IP: 128.155.10.1 Last Usable IP: 128.155.10.126 Broadcast Address: 128.155.10.127 Subnet mask: 255.255.255.128

						Subnet2: Network Address: 128.155.10.128 First Usable IP: 128.155.10.129 Last Usable IP: 128.155.10.254 Broadcast Address: 128.155.10.255 Subnet mask: 255.255.255.128
Devices For Marketing Department	128.155.10.11 To 128.155.10.50	255.255.255.128	128.155.10.1	DHCP	VLAN 10	Subnet1: Network Address: 128.155.10.0 First Usable IP: 128.155.10.1 Last Usable IP: 128.155.10.126 Broadcast Address: 128.155.10.127 Subnet mask: 255.255.255.128 Subnet2: Network Address: 128.155.10.128 First Usable IP: 128.155.10.129 Last Usable IP: 128.155.10.254 Broadcast Address: 128.155.10.255 Subnet mask: 255.255.255.128
Devices from Marketing Department	128.155.10.51 To 128.155.10.90	255.255.255.128	128.155.10.1	DHCP	VLAN 10	Subnet1: Network Address: 128.155.10.0 First Usable IP: 128.155.10.1 Last Usable IP: 128.155.10.126 Broadcast Address: 128.155.10.127 Subnet mask: 255.255.255.128 Subnet2: Network Address: 128.155.10.128 First Usable IP: 128.155.10.129 Last Usable IP: 128.155.10.254 Broadcast Address: 128.155.10.255

						Subnet mask: 255.255.255.128
Floor 2						
Managed POE switch At public area floor2	128.155.0.3	255.255.255.0	128.155.0.1	Static	none	none
Devices from IT Department	128.155.20.2	255.255.255.192	128.155.20.1	Static	VLAN 20	Subnet1: Network Address: 128.155.20.0 First Usable IP: 128.155.20.1 Last Usable IP: 128.155.20.62 Broadcast Address: 128.155.20.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.20.64 First Usable IP: 128.155.20.65 Last Usable IP: 128.155.20.126 Broadcast Address: 128.155.20.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.20.128 First Usable IP: 128.155.20.129 Last Usable IP: 128.155.20.190 Broadcast Address: 128.155.20.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.20.192 First Usable IP: 128.155.20.193 Last Usable IP: 128.155.20.254

						Broadcast Address: 128.155.20.255 Subnet mask: 255.255.255.192
Device from IT Department	128.155.21.2	255.255.255.192	128.155.21.1	Static	VLAN 21	Subnet1: Network Address:128.155.21.0 First Usable IP: 128.155.21.1 Last Usable IP: 128.155.21.62 Broadcast Address: 128.155.21.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.21.64 First Usable IP: 128.155.21.65 Last Usable IP: 128.155.21.126 Broadcast Address: 128.155.21.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.21.128 First Usable IP: 128.155.21.129 Last Usable IP: 128.155.21.190 Broadcast Address: 128.155.21.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.21.192 First Usable IP: 128.155.21.193 Last Usable IP: 128.155.21.254 Broadcast Address: 128.155.21.255 Subnet mask: 255.255.255.192
Device from HR Department	128.155.22.2	255.255.255.192	128.155.22.1	Static	VLAN 22	Subnet1: Network Address: 128.155.22.0

						First Usable IP: 128.155.22.1 Last Usable IP: 128.155.22.62 Broadcast Address: 128.155.22.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.22.64 First Usable IP: 128.155.22.65 Last Usable IP: 128.155.22.126 Broadcast Address: 128.155.22.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.22.128 First Usable IP: 128.155.22.129 Last Usable IP: 128.155.22.190 Broadcast Address:1 28.155.22.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.22.192 First Usable IP:128.155.22.193 Last Usable IP: 128.155.22.254 Broadcast Address: 128.155.22.255 Subnet mask: 255.255.255.192
One Printer In Public Area Floor 2	128.155.23.2	255.255.255.192	128.155.23.1	Static	VLAN 23	Subnet1: Network Address: 128.155.23.0 First Usable IP: 128.155.23.1 Last Usable IP: 128.155.23.62 Broadcast Address: 128.155.23.63 Subnet mask: 255.255.255.192

						Subnet2: Network Address: 128.155.23.64 First Usable IP: 128.155.23.65 Last Usable IP: 128.155.23.126 Broadcast Address: 128.155.23.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.23.128 First Usable IP: 128.155.23.129 Last Usable IP: 128.155.23.190 Broadcast Address: 128.155.23.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.23.192 First Usable IP:128.155.23.193 Last Usable IP: 128.155.23.254 Broadcast Address:128.155.23.255 Subnet mask: 255.255.255.192
U6pro Access Point In Floor 2	128.155.24.2	255.255.255.128	128.155.24.1	Static	VLAN 24	Subnet1: Network Address:128.155.24.0 First Usable IP:128.155.24.1 Last Usable IP:128.155.24.126 Broadcast Address:128.155.24.127 Subnet mask: 255.255.255.128 Subnet2: Network Address:128.155.24.128 First Usable IP:128.155.24.129

						Last Usable IP:128.155.24.254 Broadcast Address:128.155.24.255 Subnet mask: 255.255.255.128
Devices from HR Department	128.155.24.11 To 128.155.24.40	255.255.255.128	128.155.24.1	DHCP	VLAN 24	Subnet1: Network Address: 128.155.24.0 First Usable IP: 128.155.24.1 Last Usable IP: 128.155.24.126 Broadcast Address: 128.155.24.127 Subnet mask: 255.255.255.128 Subnet2: Network Address: 128.155.24.128 First Usable IP: 128.155.24.129 Last Usable IP: 128.155.24.254 Broadcast Address: 128.155.24.255 Subnet mask: 255.255.255.128
Devices from Legal Department	128.155.24.41 To 128.155.24.70	255.255.255.128	128.155.24.1	DHCP	VLAN25	Subnet1: Network Address: 128.155.24.0 First Usable IP: 128.155.24.1 Last Usable IP: 128.155.24.126 Broadcast Address: 128.155.24.127 Subnet mask: 255.255.255.128 Subnet2: Network Address: 128.155.24.128 First Usable IP: 128.155.24.129 Last Usable IP: 128.155.24.254 Broadcast Address: 128.155.24.255
Floor 3						
Managed POE switch In Floor 3	128.155.0.4	255.255.255.0	128.155.0.1	Static	None	none

Devices from Customer Service Department	128.155.30.2	255.255.255.192	128.155.30.1	Static	VLAN30	Subnet1: Network Address: 128.155.30.0 First Usable IP: 128.155.30.1 Last Usable IP: 128.155.30.62 Broadcast Address: 128.155.30.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.30.64 First Usable IP: 128.155.30.65 Last Usable IP: 128.155.30.126 Broadcast Address: 128.155.30.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.30.128 First Usable IP: 128.155.30.129 Last Usable IP:128.155.30.190 Broadcast Address: 128.155.30.191 Subnet mask: 255.255.255.192 Subnet4: Network Address:1 28.155.30.192 First Usable IP: 128.155.30.193 Last Usable IP: 128.155.30.254 Broadcast Address: 128.155.30.255 Subnet mask: 255.255.255.192
Devices from Customer Service Department	128.155.31.2	255.255.255.192	128.155.31.1	Static	VLAN31	Subnet1: Network Address: 128.155.31.0 First Usable IP: 128.155.31.1 Last Usable IP: 128.155.31.62

						Broadcast Address: 128.155.31.63 Subnet mask: 255.255.255.192 Subnet2: Network Address:128.155.31.64 First Usable IP:128.155.31.65 Last Usable IP:128.155.31.126 Broadcast Address:128.155.31.127 Subnet mask:255.255.255.192 Subnet3: Network Address:128.155.31.128 First Usable IP:128.155.31.129 Last Usable IP:128.155.31.190 Broadcast Address:128.155.31.191 Subnet mask:255.255.255.192 Subnet4: Network Address:128.155.31.192 First Usable IP:128.155.31.193 Last Usable IP:128.155.31.254 Broadcast Address:128.155.31.255 Subnet mask:255.255.255.192
Device From Product Department	128.155.32.2	255.255.255.192	128.155.32.1	Static	VLAN32	Subnet1: Network Address: 128.155.32.0 First Usable IP: 128.155.32.1 Last Usable IP: 128.155.32.62 Broadcast Address: 128.155.32.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.32.64

						First Usable IP: 128.155.32.65 Last Usable IP: 128.155.32.126 Broadcast Address: 128.155.32.127 Subnet mask: 255.255.255.192 Subnet3: Network Address: 128.155.32.128 First Usable IP: 128.155.32.129 Last Usable IP: 128.155.32.190 Broadcast Address: 128.155.32.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.32.192 First Usable IP: 128.155.32.193 Last Usable IP: 128.155.32.254 Broadcast Address: 128.155.32.255 Subnet mask: 255.255.255.192
One Printer In Floor 3	128.155.33.2	255.255.255.192	128.155.33.1	Static	VLAN33	Subnet1: Network Address: 128.155.33.0 First Usable IP: 128.155.33.1 Last Usable IP: 128.155.33.62 Broadcast Address: 128.155.33.63 Subnet mask: 255.255.255.192 Subnet2: Network Address: 128.155.33.64 First Usable IP: 128.155.33.65 Last Usable IP: 128.155.33.126 Broadcast Address: 128.155.33.127 Subnet mask: 255.255.255.192

						Subnet3: Network Address: 128.155.33.128 First Usable IP: 128.155.33.129 Last Usable IP: 128.155.33.190 Broadcast Address: 128.155.33.191 Subnet mask: 255.255.255.192 Subnet4: Network Address: 128.155.33.192 First Usable IP: 128.155.33.193 Last Usable IP: 128.155.33.254 Broadcast Address: 128.155.33.255 Subnet mask: 255.255.255.192
U6pro Access Point in Floor 3	128.155.34.2	255.255.255.128	128.155.34.1	Static	VLAN34	Subnet1: Network Address: 128.155.34.0 First Usable IP: 128.155.34.1 Last Usable IP: 128.155.34.126 Broadcast Address: 128.155.34.127 Subnet2: Network Address: 128.155.34.128 First Usable IP: 128.155.34.129 Last Usable IP: 128.155.34.254 Broadcast Address: 128.155.34.255
Device from Operation Department	128.155.34.11 To 128.155.34.40	255.255.255.128	128.155.34.1	DHCP	VLAN34	Subnet1: Network Address: 128.155.34.0 First Usable IP: 128.155.34.1 Last Usable IP: 128.155.34.126 Broadcast Address: 128.155.34.127

						Subnet2: Network Address: 128.155.34.128 First Usable IP: 128.155.34.129 Last Usable IP: 128.155.34.254 Broadcast Address: 128.155.34.255
Device from Operation Department	128.155.34.41 To 128.155.34.70	255.255.255.128	128.155.34.1	DHCP	VLAN34	Subnet1: Network Address: 128.155.34.0 First Usable IP: 128.155.34.1 Last Usable IP: 128.155.34.126 Broadcast Address: 128.155.34.127 Subnet2: Network Address: 128.155.34.128 First Usable IP: 128.155.34.129 Last Usable IP: 128.155.34.254 Broadcast Address: 128.155.34.255

3.0 Error Detection Mechanism

To ensure data being transferred correctly and successfully, it is important to implement error detection mechanism so that error can be detect when the receiver side fails to receive the data or the data is corrupted. This is important so that actions can be carried out to solve the error. There are 3 types of error detection mechanism which is Parity Check, Cyclic Redundancy Check (CRC) and Checksum. For this scenario, CRC and Checksum is being used as Parity Check cannot detect multiple errors (Eureka by patsnap, 2025).

3.1 Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check is a mathematical method where it operates by using polynomial divisions. First, the sender side will add a few bits of zero at the end of the bits in the original data. Then, the bits of data will be divided into the generator polynomial. The remaining of the number is known as the CRC checksum. The checksum and the original data is then combined and sent to the receiver side. The receiver side will then repeat the same steps. If the remainder is 0, it means the data is not corrupted. If it is not 0, then the data is corrupted and actions should be taken (Kumar, 2026).

In this proposed network, CRC is used at the access points where the staff of the company use it during video conferencing and accessing cloud applications (reverspecb, 2023). This is important for seamless user experience as packet loss could cause the video to freeze and dropped words in the audio during video conferencing. Besides this, loosing data packets when using cloud application could cause the performance of the cloud application to degrade as delays could happened (cyfuture cloud, n.d.). These could cause frustration amongst staff.

3.2 Checksum

Checksum is a mechanism that runs by a mathematical method. First, the data will be divided into several segments and each of them will have the same bits. These segments will then be arranged into an orderly sequence. For example, we segment it into 5 parts which is a, b, c, d, e from the original data. First, we will add the binary for segment a and b together and add 1 to the binary result. Then, the calculation will be repeated until segment e. Next, invert the result, which is switching the 1s to 0s and the 0s to 1s. Then, the result will be appended to the original data and send to the receiver. The receiver will then repeat the process that had been done by the sender. If the result is all 0, it means that data is correct, else if there exist any 1 bit inside the result, it indicates that the data is corrupted. (Hussaini, 2022)

In this proposed network, checksum is being used during downloading large folder. This happens as the receiver side will run the calculation to ensure the folder is downloaded fully where none of the data or bytes had gone missing. If the result of the checksum is wrong, it means that the file downloaded is ideally wrong or malicious had replaced the original data. This is important for the company as it can prevent malicious files and bugs from controlling the company (ManageEngine, n.d.).

4.0 Scalability and Security

4.1 Scalability

Scalability is important factor to consider in a network design since it should accommodate future growth of the company.

In the proposed network diagram, the modular core switch plays a vital role in scalability. If the existing port for the modular core switch is not enough. The company can configure the port flexibly. This happens as expansion of the modular core switch can be done through addition of modules. (Powell, 2025).

Besides modular core switch, the router also helps in scalability. The router includes Software-Defined Wide Area Network (SD-WAN) where it has the characteristic of scalability. SD-WAN supports a wide range of connectivity options causing it to be able to adapt different environment and needs flexibly. This cause SD-WAN to be able to evolve with the company business and scale the network up or down according to the needs of the company (Fortinet , n.d.).

Additionally, UniFi Application Suite also helps in scalability as it contains auto-scale network where it automatically adjusts network settings according to the usage patterns. When the system thinks that the IP address is running out in the DHCP range and subnet mask, it will expand to a larger subnet mask to provide more IP to the company automatically (Howe, 2025).

All in all, modular core switch, SD-WAN in router and auto-scale network in UniFi Application Suite collaborates with each other to provide scalability to the network system.

4.2 Security

Security is always an aspect that is important in network design as it helps to prevent loss from happening. In this proposed network design, the router that is recommended includes IDS and IPS. These two systems are important as it will help to monitor the network traffic and search for any suspicious or malicious activity going on. When a threat is found, IDS would alert the company while IPS will take proactive steps to block the threats that had been detected.

Besides this, the router also includes Deep Packet Inspection (DPI) where it is also known as packet sniffing. DPI will analyse the packets to detect hidden threats. This is important as it can helps to strengthen the security of the network as encrypted attacks and hidden malware can be detected (Network Critical , n.d.).

Moreover, the router also includes a stateful firewall and an Application-Aware Layer 7 Firewall. A stateful firewall is a firewall that helps to track and monitor the state of active network connections during analysing incoming traffic and finding for potential traffic and data risks. This firewall is important as it can detect the attempts of unauthorize individuals trying to access to the network (Fortinet home, n.d.). As for the Application Layer 7 firewall, it will intercept the network traffic destined for or originating from protected application and identify protocol being used. Then, it examines the content of packet by checking for its vulnerabilities, malware signature and other threats. Based on the result and policies that had been configured by the company, the firewall will take appropriate actions whether to block the traffic, allow the traffic, alert administrators and may more (Powell, 2025).

All in all, the router includes most of the security needed to prevent security breaches from happening.

5.0 Conclusion

In conclusion, the proposed network configuration is suitable for the 3-storey office block building with approximately 100 employees in the 3 units of the company. The UDM Pro router, module-based core switch, managed PoE switches, the UniFi U6 Pro access points, Cat6 FTP cables and 10G SFP+ DAC cables support high internet usage, video conferencing and cloud-based applications. VLANs help organize the traffic and improve network management while IP addressing scheme helps in controlling network traffic, and CRC and checksum functions help to make sure data transmission is accurate. The network also includes security measures like IDS, IPS, Deep Packet Inspection (DPI) and firewalls, which are designed to safeguard against unauthorized access and malicious activity. Additionally, the modular core switch and wireless access points provide scalability, allowing the network to expand as the business grows, without impacting network performance.

6.0 Appendix

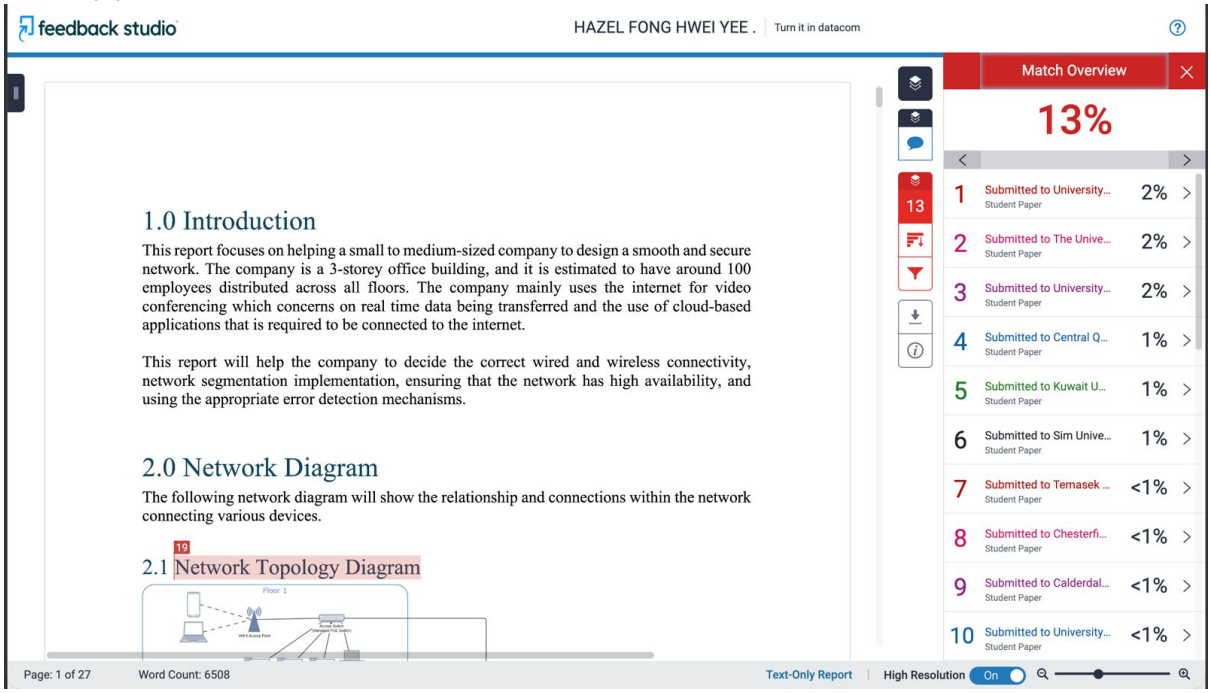


Figure 9 Turn it in

7.0 References

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